

Challenge (Habanero)

- Q1.** In a gaming tournament, the number of wins x and the average score y are related. If the data points are $(1, 50)$, $(2, 60)$, $(3, 70)$, what type of correlation exists?
(A) Positive
(B) Negative
(C) None
(D) Can't determine
(E) Insufficient data
- Q2.** The line of best fit for the data set $(2, 4)$, $(4, 6)$, $(6, 8)$ is $y =$
(A) $x + 1$
(B) $2x - 2$
(C) $x - 2$
(D) x
(E) $x + 2$
- Q3.** A gamer's score y and the time x spent playing are modeled by the equation $y = 2x + 10$. What is the y -intercept?
(A) 2
(B) 10
(C) 0
(D) -2
(E) -10
- Q4.** If a line of best fit has a slope of 3 and passes through the point $(1, 2)$, its equation is $y =$
(A) $3x - 1$
(B) $3x + 1$
(C) $3x - 2$
(D) $2x + 1$
(E) $x + 3$
- Q5.** The data points $(0, 0)$, $(1, 2)$, $(2, 4)$ have a line of best fit with a slope of $m =$
(A) 2
(B) 1
(C) 0
(D) -1
(E) -2
- Q6.** If a scatter plot shows no correlation, what does the line of best fit do?
(A) Has a slope of 1
(B) Is a horizontal line
(C) Is a vertical line
(D) Passes through the origin
(E) Does not exist
- Q7.** In a coding competition, the line of best fit for the number of lines x and time y taken is $y = 0.5x + 1$. Predict the time for 10 lines of code.
(A) 4
(B) 5
(C) 6
(D) 5.5
(E) 6.5
- Q8.** The graph of $y = 2x - 3$ is the line of best fit for the data set $(1, 1)$, $(2, 3)$, $(3, 5)$. What is the y -intercept?
(A) 1
(B) -3
(C) 3
(D) -1
(E) 2

Open Ended Questions (FRQ)

- Q9.** A gamer's score y and the number of hours x spent playing are related. The data points are $(1, 20)$, $(2, 40)$, $(3, 60)$. Draw the line of best fit and use it to predict the score for 4 hours of playing. Show your work and explain your reasoning. Provide the equation of the line of best fit.
- Q10.** The number of wins x and the average score y in a gaming tournament are related. The data set is $(2, 30)$, $(4, 50)$, $(6, 70)$. Create a scatter plot, identify the type of correlation, and draw the line of best fit. Use the line of best fit to predict the average score for 8 wins. Show your work and provide the equation of the line of best fit.

Chef's Tips

- Provide clear instructions and examples before the test.
- Ensure students have the necessary tools and materials.
- Encourage students to ask questions and seek help when needed.